1 Classifier(-free) Guidance

Discussed on 25 Aug 2025

1.1 Original CFG Paper

Tim Salimans Jonathan Ho (2021). "Classifier-Free Diffusion Guidance". In: *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*

- Implementing truncation or low temperature sampling in diffusion models is ineffective. Scaling model scores or decreasing the variance of Gaussian noise of the reverse process generates blurry, low quality samples (diffusion models beat GANs paper). It is therefore not trivial how to influence the sampling process of a diffusion model.

- Vanilla CG is a way to trade off IS and FID by "up-weighting the probability of data for which the classifier assigns high likelihood to the correct label". CG adapts the learned score $\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) \approx -\sigma_\lambda \nabla_{\mathbf{z}_\lambda} \log p(\mathbf{z}_\lambda \mid \mathbf{c})$ to include the gradient of the log likelihood of an auxiliary classifier model $p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)$ as follows:

$$\begin{aligned}\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_\lambda, \mathbf{c}) &= \boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - w\sigma_\lambda \nabla_{\mathbf{z}_\lambda} \log p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda) \\ &\approx -\sigma_\lambda \nabla_{\mathbf{z}_\lambda}\left[\log p(\mathbf{z}_\lambda \mid \mathbf{c}) + w \log p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)\right]\end{aligned}$$

  This results in approximate samples from the distribution:

$$\tilde{p}_\theta(\mathbf{z}_\lambda \mid \mathbf{c}) \propto p_\theta(\mathbf{z}_\lambda \mid \mathbf{c})\, p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)^w$$

  Results on a toy 2D dataset with 3 classes for which the conditional distribution for each class is an isotropic Gaussian show that the conditional upon applying guidance is non-Gaussian. As guidance strength increases, each conditional places probability mass farther away from other classes and towards directions of high confidence given by logistic regression, and most of the mass becomes concentrated in smaller regions.

- CFG aims to achieve the same modification of the conditional score but without a classifier. It is done by training a conditional and unconditional model with one neural network and trade off both scores during sampling

$$\tilde{\boldsymbol{\epsilon}}_\theta(\mathbf{z}_\lambda, \mathbf{c}) = (1+w)\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda, \mathbf{c}) - w\boldsymbol{\epsilon}_\theta(\mathbf{z}_\lambda)$$

  Only a relatively small portion of the model capacity of the diffusion model needs to be dedicated to the unconditional generation task in order to produce classifier-free guided scores effective for sample quality.

- CFG extends vanilla classifier guidance to trade off mode coverage and sample fidelity in conditional diffusion model sampling in analogy to low temperature sampling or truncation in other types of generative models.

- CFG may be better than vanilla classifier guidance could be seen as an adversarial attack on classifiers, which could boost classifier-based metrics like FID and IS

Algorithm 1 Conditional sampling with classifier-free guidance

Require: w guidance strength
Require: c conditioning information for conditional sampling
Require: $\lambda_1, \ldots, \lambda_T$: increasing log SNR sequence with $\lambda_1 = \lambda_{\min}, \lambda_T = \lambda_{\max}$
  $\mathbf{z}_1 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  for $t = 1, \ldots, T$ **do**
    $\tilde{\boldsymbol{\epsilon}}_t = (1+w)\boldsymbol{\epsilon}_\theta(\mathbf{z}_t, \mathbf{c}) - w\boldsymbol{\epsilon}_\theta(\mathbf{z}_t)$ ▷ CFG score at log SNR λ_t
    $\tilde{\mathbf{x}}_t = (\mathbf{z}_t - \sigma_{\lambda_t}\tilde{\boldsymbol{\epsilon}}_t)/\alpha_{\lambda_t}$ ▷ Denoising step based on sampler
    if $t < T$ **then**
      $\mathbf{z}_{t+1} \sim \mathcal{N}\left(\tilde{\boldsymbol{\mu}}_{\lambda_{t+1}|\lambda_t}(\mathbf{z}_t, \tilde{\mathbf{x}}_t), \left(\tilde{\sigma}^2_{\lambda_{t+1}|\lambda_t}\right)^{1-v}\left(\sigma^2_{\lambda_t|\lambda_{t+1}}\right)^v\right)$
    else
      $\mathbf{z}_{t+1} = \tilde{\mathbf{x}}_t$
    end if
  end for

Questions

- Trade-off between diversity and sample quality (truncation in GAN, temperature sampling in flows, cfg in diffusion). Intuition why these two things are opposing?
- People usually discuss diversity and sample quality because it can be measured by IS and FID, but what about the success of the conditioning (prompt alignment). Is that always given or are there also degrees to it? It should not be the same to barely cross the class boundary compared to sample in the mode of the target class?
- In the 2D toy example, why is the distribution changed in such a way? First, why is mass shifted away from other classes even though the gradient is not explicitly calculated for it (or is that because of the low dimensionality)? Second, why is the mass concentrated (in a bit more complicated class, there should be multiple high-likelihood regions)?
- "Furthermore, $\tilde{\boldsymbol{\epsilon}}_\theta$ is constructed from score estimates that are non-conservative vector fields due to the use of unconstrained neural networks, so there in general cannot exist a scalar potential such as a classifier log likelihood for which $\tilde{\boldsymbol{\epsilon}}_\theta$ is the classifier-guided score." What does that mean?
- CFG is trained to mostly be a conditional model and then the conditional score is again heavily amplified during sampling. How is that different to a conditional model (with regularization)?

- Both CG and CFG obtain their best results when applying guidance to an already class-conditional model, as opposed to applying guidance to an unconditional model. In theory, they are proportional as $p_\theta(\mathbf{z}_\lambda \mid \mathbf{c})\, p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)^w \propto p_\theta(\mathbf{z}_\lambda)\, p_\theta(\mathbf{c} \mid \mathbf{z}_\lambda)^{w+1}$. Is guidance on the conditional model simply better because it is easier to regularize the class-conditional model?

1.2 (skip) Elucidating the Design Space of Diffusion-Based Generative Models

Tero Karras et al. (2022). "Elucidating the design space of diffusion-based generative models". In: *NeurIPS*

- The choice of the noise schedule has major practical implications and should not be made on the basis of theoretical convenience.

- Heun's 2nd order method (a.k.a. improved Euler, trapezoidal rule) — previously explored in the context of diffusion models by Jolicoeur-Martineau et al. — provides an excellent tradeoff between truncation error and NFE. It introduces an additional correction step for xi+1 to account for change in dx/dt between ti and ti+1. This correction leads to less local error at the cost of one additional evaluation of $D_\theta(\boldsymbol{x}; t)$ per step.

- The time steps determine how the step sizes and thus ODE truncation errors are distributed between different noise levels. The step size should decrease monotonically with decreasing noise schedule, and it does not need to vary on a per-sample basis.

- The shape of the ODE solution trajectories is defined by the noise schedule and noise scale functions. The choice of these functions offers a way to reduce the truncation errors, as their magnitude can be expected to scale proportional to the curvature of dx/dt. The best choice for these functions is $\sigma(t) = t$ (noise schedule) and $s(t) = 1$ (noise scale), which is also the choice made in DDIM. With this choice, the ODE simplifies to $\mathrm{d}\boldsymbol{x}/\mathrm{d}t = (\boldsymbol{x} - D(\boldsymbol{x}; t))/t$ and σ and t become interchangeable. An immediate consequence is that at any x and t, a single Euler step to $t = 0$ yields the denoised image $D_\theta(\boldsymbol{x}; t)$. The tangent of the solution trajectory therefore always points towards the denoiser output. This can be expected to change only slowly with the noise level, which corresponds to largely linear solution trajectories.

- The sampling process is orthogonal to how each model was originally trained.

- The SDEs of Song et al. can be generalized as a sum of the probability flow ODE and a time-varying Langevin diffusion SDE:

$$\mathrm{d}\boldsymbol{x}_\pm = \underbrace{-\dot\sigma(t)\sigma(t)\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}; \sigma(t))\mathrm{d}t}_{\text{probability flow ODE (Eq. 1)}} \pm \underbrace{\underbrace{\beta(t)\sigma(t)^2\nabla_{\boldsymbol{x}} \log p(\boldsymbol{x}; \sigma(t))\mathrm{d}t}_{\text{deterministic noise decay}} + \underbrace{\sqrt{2\beta(t)}\sigma(t)\mathrm{d}\omega_t}_{\text{noise injection}}}_{\text{Langevin diffusion SDE}},$$

where ω_t is the standard Wiener process. dx+ and dx- are now separate SDEs for moving forward and backward in time, related by the time reversal formula of Anderson. The Langevin term can further be seen as a combination of a deterministic score-based denoising term and a stochastic noise injection term, whose net noise level contributions cancel out. As such, $\beta(t)$ effectively expresses the relative rate at which existing noise is replaced with new noise. This perspective reveals why stochasticity is helpful in practice: The implicit Langevin diffusion drives the sample towards the desired marginal distribution at a given time, actively correcting for any errors made in earlier sampling steps. On the other hand, approximating the Langevin term with discrete SDE solver steps introduces error in itself. The optimal amount of stochasticity should be determined empirically, however excessive Langevin-like addition and removal of noise results in gradual loss of detail in the generated images with all datasets and denoiser networks. There is also a drift toward oversaturated colors at very low and high noise levels. Interestingly, the relevance of stochastic sampling appears to diminish as the model itself improves.

1.3 (skip) Entropy-driven sampling and training scheme for conditional diffusion generation

Guangcong Zheng et al. (2022). "Entropy-driven sampling and training scheme for conditional diffusion generation". In: *ECCV*

- Due to the gap between the discriminative task of the classifier and the generative task of the diffusion model, the classifier can easily classify the images in the middle of the sampling process. As a result, the guidance vanishes and the details are left for the diffusion model.

- To avoid this, the guidance scale can be rescaled adaptively based on the entropy of the predicted classifier class distribution. The scale is calculated based on the predicted distribution entropy normalized by the maximum entropy (entropy of the uniform distribution over the classes). This has been extended to a training adaption.

1.4 (skip) Generating high fidelity data from low-density regions using diffusion models

Vikash Sehwag et al. (2022). "Generating high fidelity data from low-density regions using diffusion models". In: *CVPR*

- Vanilla sampling from diffusion models predominantly sample from high-density regions of the data manifold.

- By using the vanilla classifier guidance setup, the denoising process is steered by two additional guidance terms: 1) minimizing a contrastive loss for a Gaussian model of image embeddings and class embeddings in the embedding space of a classifier, which pushed the sample towards low density regions and 2) a binary classifier to distinguish fake and real images to generate more images on the manifold, as higher guidance scales or when scaling term 1) too much leads to off-manifold generations.

- The high likelihood of a sample could mean a sample from a higher density region, but the likelihood estimates don't seem to be a reliable predictor of manifold density by common metrics and human judgement.
- Diffusion models seem to generalize in low density regions of the data manifold instead of relying on memorization, at least on an imagenet scale.

1.5 Characteristic Guidance: Non-linear Correction for Diffusion Model at Large Guidance Scale

Candi Zheng and Yuan Lan (2023). "Characteristic guidance: Non-linear correction for diffusion model at large guidance scale". In: *ICML*

- The original CFG formulation relies on the mixing of scores for the conditional and unconditional model, but the authors say this only holds for $t = 0$. The authors propose a mixing error, which measures the deviation of the guidance method from the Fokker-Planck equation (the learned score function is a solution of the score FP equation, the deviation from this leads to errors in sampling due to removing the equivalence between forward and backward diffusion process). For CFG, the mixing error comes from the non-linear terms of the FP equation, where the amplitude escalates with an increase in guidance scale and intensifies as the variance decreases.
- Characteristic guidance is proposed as:

$$\boldsymbol{\epsilon}_{CH}\left(\mathbf{x} \mid \mathbf{c}, t_i, \omega\right) = (1+\omega)\boldsymbol{\epsilon}_\theta\left(\mathbf{x}_1 \mid \mathbf{c}, t_i\right) - \omega\boldsymbol{\epsilon}_\theta\left(\mathbf{x}_2, t_i\right) \quad ,$$

  with $\mathbf{x}_1 = \mathbf{x} + \omega\Delta\mathbf{x}, \mathbf{x}_2 = \mathbf{x} + (1+\omega)\Delta\mathbf{x}$ and $\Delta\mathbf{x}$ a training-free non-linear correction term based on a projection operation. This projection is different for pixel space diffusion, latent space diffusion and low dimensional cases.
- The paper proposes characteristic guidance as a systematic solution to the large guidance scale issue by addressing the nonlinearity neglected in classifier-free guidance.
- The non-linear correction predominantly occurs in the middle stages of the diffusion process.

Questions

- The authors claim that CFG suffers from severe bias and catastrophic loss of diversity when ODE based sampling methods (ODE, DDIM, and DPM++2M) are used. Why is that the case?
- The characteristic guidances effect on latent space diffusion models (LDM) (Rombach et al., 2021) is different: it enhances the control strength rather than lowering the FID. It may simply be the choice of projection but its not clear why.

1.6 (skip) Spontaneous Symmetry Breaking in Generative Diffusion Models

Gabriel Raya and Luca Ambrogioni (2023). "Spontaneous symmetry breaking in generative diffusion models". In: *NeurIPS*

- The paper proposes that the generative process in diffusion models consists of two phases 1) A linear steady-state dynamics around a central fixed-point and 2) an attractor dynamics directed towards the data manifold. These two "phases" are separated by the change in stability of the central fixed-point, with the resulting window of instability being responsible for the diversity of the generated samples.
- The theory is empirically shown on some data sets, but leveraging symmetries in practice is very difficult due to the high dimensionality

1.7 (skip) Universal Guidance for Diffusion Models

Arpit Bansal et al. (2023). "Universal Guidance for Diffusion Models". In: *CVPR Workshops*

- Universal Guidance is a heuristic improvement over the vanilla classifier guidance based on three steps: forward universal guidance, backward universal guidance and per-step self-recurrence
- Forward universal guidance is applying a one-step denoising estimate, which is then used to guide the unconditional noise prediction for the noisy latent
- Backward universal guidance optimizes over the one-step estimate (in data space) for maximum conditioning alignment, then adapts the noise prediction for the noisy latent by the delta in the data space
- Per-step self-recurrence moves one step towards noise (against the sampling direction) and repeats all three steps k times

1.8 (skip) Rethinking conditional diffusion sampling with progressive guidance

Anh-Dung Dinh et al. (2023). “Rethinking conditional diffusion sampling with progressive guidance”. In: *NeurIPS*

- As classifier gradients wrt to one class try to push the generated sample away from other classes, common features across classes may be suppressed. This results in a limited pool of features with less diversity or the absence of robust features for image construction.
- Instead of maximizing the probability towards one class, one can guide by the weighted sum over all class gradients. The gradients are weighted by LLM-generated text descriptions and the resulting embedding similarity. The main target class is also progressively upscaled whereas the influence of the other classes is downscaled accordingly throughout the sampling process.

1.9 (skip) Freedom: Training-free energy-guided conditional diffusion model

Jiwen Yu et al. (2023). “Freedom: Training-free energy-guided conditional diffusion model”. In: *ICCV*

- Guidance should be training-free, meaning no extra robust classifier training (CG) and no conditional score estimation (CFG)
- Model the conditional alignment distribution as:

  $$p\left(\mathbf{c} \mid \mathbf{x}_t\right)=\frac{\exp \left\{-\lambda \mathcal{E}\left(\mathbf{c}, \mathbf{x}_t\right)\right\}}{Z} \quad ,$$

  s.t. $\nabla_{\mathbf{x}_t} \log p\left(\mathbf{c} \mid \mathbf{x}_t\right) \propto -\nabla_{\mathbf{x}_t} \mathcal{E}\left(\mathbf{c}, \mathbf{x}_t\right)$
- In practice, the energy is a distance function for an arbitrary conditioning and a one step denoising estimate is used to apply the conditioning.
- As suggested in previous work, guidance is most impactful in the middle of the sampling process, where semantics are introduced. The authors propose to apply the energy guidance in that interval and further amplify the guidance by ”time-travel”, which noises x_t multiple time steps and guides back to t, effectively shifting more compute and conditioning in the middle of the sampling process.

1.10 (skip) Loss-Guided Diffusion Models for Plug-and-Play Controllable Generation

Jiaming Song et al. (2023). “Loss-Guided Diffusion Models for Plug-and-Play Controllable Generation”. In: *PMLR* Use classifier guidance, but estimate $p_t(y \mid$

x_t) as:

$$p_t^{(\ell)}(\mathbf{y} \mid \mathbf{x}_t) = \int_{\mathbf{x}_0} p(\mathbf{x}_0 \mid \mathbf{x}_t)\, p_0^{(\ell)}(\mathbf{y} \mid \mathbf{x}_0)\, \mathrm{d}\mathbf{x}_0$$

$p(\mathbf{x}_0 \mid \mathbf{x}_t)$ requires multiple model evaluations, so it is approximated as:

$$\mathbf{MC}_n(\mathbf{x}_t, \mathbf{y}) = \nabla_{\mathbf{x}_t} \log\left(\frac{1}{n}\sum_{i=1}^{n} \exp\left(-\ell_{\mathbf{y}}\left(\mathbf{x}^{(i)}\right)\right)\right) \quad ,$$

evaluated $n = 3$ times during sampling, where $\mathbf{x}^{(i)} \sim q(\mathbf{x}_0 \mid \mathbf{x}_t)$ are i.i.d. samples from an approximate inference distribution $q(\mathbf{x}_0 \mid \mathbf{x}_t)$. In practice, it's a Gaussian centered around the minimum mean squared error (MMSE) estimator of x_0 given x_t and σ_t (basically optimal x_o prediction).

1.11 Boosting Diffusion Models with an Adaptive Momentum Sampler

Xiyu Wang et al. (2024b). "Boosting Diffusion Models with an Adaptive Momentum Sampler." In: *IJCAI*

- Standard Diffusion model sampling is independent of previous time steps, which can lead to excessive noise and missing high-level components in the generated images.
- By using an ADAM-like sampler and employing a larger momentum that heavily re-uses the update direction of early steps leads to augmented details while still preserving high-level information. This phenomenon aligns with the two stages of learning for likelihood-based models (Rombach et al. 2022), where the first stage focuses on learning high-frequency details, while the second stage focuses on learning conceptual compositions.

Questions

- "IS (...) is often referred to as a method to measure inter-class diversity and is less sensitive to the diversity of the images inside one label (Lucic et al. 2018; Borji 2019). In contrast, FID can detect intra-class mode collapsing as well (Lucic et al. 2018)." Does that mean that the blind spot of both metrics are non diverse intra-class samples which are however not a mode collapse? Is that even desired?
- The authors show a rate (bits/dim) vs distortion (RMSE) curve. What does that mean?

1.12 (skip) Theoretical insights for diffusion guidance: a case study for Gaussian mixture models

Yuchen Wu et al. (2024). "Theoretical insights for diffusion guidance: a case study for Gaussian mixture models". In: *ICML*

- For GMMs and both DDPM and DDIM samplers with diffusion guidance, the classification confidence — which measures the posterior probability associated with the guided class given an output sample — only increases when diffusion guidance is applied.
- For GMMs, increasing the diffusion guidance always results in a reduction in the differential entropy. This offers the first theoretical explanation for the benefit of the diffusion guidance in generating more homogeneous samples.
- For GMMs with the means aligned, the role of the guidance strength can be more complicated. There seems to exist a phase transition in the behavior of the classification confidence as one increases the guidance strength. Cautions thus need to be exercised in practice in terms of selecting a proper guidance strength.

1.13 Classifier-Free Guidance is a Predictor-Corrector

Arwen Bradley and Preetum Nakkiran (2024). "Classifier-Free Guidance is a Predictor-Corrector". In: *Mathematics of Modern Machine Learning (M3L) Workshop at NeurIPS*

- CFG interacts differently with DDIM and DDPM, and neither sampler with CFG generates the gamma powered distribution $p(x \mid c)^{\gamma} p(x)^{1-\gamma}$
- CFG can be interpreted as a predictor-corrector methods that alternates between denoising and sharpening
- In the SDE limit, CFG is equivalent to combining a DDIM predictor for the conditional distribution with a Langevin dynamics corrector for the gamma-powered distribution $p_t(x)^{1-\gamma'} p_t(x \mid c)^{\gamma'}$, with $\gamma' = (2\gamma - 1)$
- Understanding CFG (p^* as the ideal CFG sampler, $\hat{p}$ as the real CFG sampler) - Why CFG improves both image quality and promt adherence compared to conditional sampling?:

$$\underbrace{\text{PerceivedQuality}\ [\widehat{p}_\gamma]}_{\text{Real CFG}} = \underbrace{\text{PerceivedQuality}\ [p^*_\gamma]}_{\text{Ideal CFG}} - \underbrace{\big(\text{PerceivedQuality}\ [p^*_\gamma] - \text{PerceivedQuality}\ [\widehat{p}_\gamma]\big)}_{\text{Generalization Gap}}$$

  Which means at least one of the following occurs:

  - The ideal CFG sampler improves in quality with increasing γ. That is, CFG distorts the population distribution in a favorable way (e.g. by sharpening it, or otherwise).
  - The generalization gap decreases with increasing γ. That is, CFG has a type of regularization effect, bringing population and empirical processes closer.

It is likely that both occurs: most models are trained on poor-quality datasets (cluttered images from the web) and we want the model to sample from a higher-quality distribution (images of an isolated subject). On the other hand, CFG seems to produce less outliers.

- It is open to find explicit characterizations of CFGs output distribution, in terms of the original $p(x)$ and $p(x|c)$ — although it is possible tractable expressions do not exist.

Questions

- ”We cannot mathematically specify this notion of quality, but we will assume it exists for analysis. Notably, PerceivedQuality is not a measurement of how close a distribution is to the ground-truth p(x|c) — it is possible for a generated distribution to appear even “higher quality” than the ground-truth, for example.” How?
- For the case of the reduced generalization gap, is that somehow related to the gap in the ELBO?

1.14 (skip) What does guidance do? A fine-grained analysis in a simple setting

Muthu Chidambaram et al. (2024). “What does guidance do? A fine-grained analysis in a simple setting”. In: *NeurIPS*

- The paper analyzes two cases: (1) mixtures of compactly supported distributions and (2) mixtures of Gaussians, which reflect salient properties of guidance that manifest on real-world data.
- As the guidance parameter increases, the guided model samples more heavily from the boundary of the support of the conditional distribution. The guided ODE first swings past the edges of the support of p and into the tails of the noised data distribution p_t before returning. As a result, errors in score estimation at these tails can move the sampling process away from the intended trajectory and thus prevent the trajectory from ever returning to the support of p, leading to corrupted outputs.
- For any nonzero level of score estimation error, sufficiently large guidance will result in sampling away from the support, theoretically justifying the empirical finding that large guidance results in distorted generations.

1.15 Guiding a Diffusion Model with a Bad Version of Itself

Tero Karras et al. (2024). “Guiding a diffusion model with a bad version of itself”. In: *NeurIPS*

- CFG uses an unconditional model to guide a conditional model, leading to simultaneously better prompt alignment and higher-quality images at the cost of reduced variation. Interestingly, it is possible to obtain disentangled control over image quality without compromising the amount of variation by guiding generation using a smaller, less-trained version of the model itself rather than an unconditional model.

- CFG forces the model to generate samples away from other classes, which can mean dropping density areas close to another class. CFG also avoids low-probability intermediate regions which can mean gaps in the density where CFG wont sample from. The authors say this may stem from a worse fit of the data distribution for the unconditional model due to the fact that it is much more difficult to generate out of the entire distribution, paired with the fact that the unconditional model usually gets a small part of the training budget. This could mean CFG is not only boosting the likelihood of the sample having come from the target class, but also that of having come from the higher-quality implied distribution.

- Autoguidance uses this intuition to guide a high-quality model with a poor quality model, usually with low capacity or under-trained. This is because under limited model capacity, score matching tends to over-emphasize low-probability (i.e., implausible and under-trained) regions of the data distribution. The authors expect the weaker version of the same model to make broadly similar errors in the same regions, only stronger. Measuring the difference between both scores indicates the general direction towards better samples if the models disagree. The authors recommend an early snapshot of a smaller model as the guidance model. Many other degradation models to yield a guidance model are not performing well.

Questions

- ”The training objective of a diffusion model aims to cover the entire (conditional) data distribution. This causes problems in low-probability regions: The model gets heavily penalized for not representing them, but it does not have enough data to learn to generate good images corresponding to them. CFG has become the standard method for “lowering the sampling temperature”, i.e., focusing the generation on well-learned high-probability regions. By training a denoiser network to operate in both conditional and unconditional setting, the sampling process can be steered away from the unconditional result — in effect, the unconditional generation task specifies a result to avoid.”. Should the training objective then not be fixed to ignore low probability regions from the start?

1.16 Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models

Tuomas Kynkäänniemi et al. (2024). "Applying Guidance in a Limited Interval Improves Sample and Distribution Quality in Diffusion Models". In: *NeurIPS*

- The authors show that guidance is clearly harmful toward the beginning of the backward process (high noise levels) leading towards a handful template images per prompt greatly reducing variation, largely unnecessary toward the end (low noise levels), and only beneficial in the middle as it causes samples with more decisive set of features with crisper and perceptually more pleasing results.
- Limiting the guidance interval allows the use of much higher guidance weight.
- The proposed guidance interval is based on a piecewise-linear function, which sets w to 1 outside the guidance interval, removing the influence of the unconditional model. Smoother weighing functions did not improve results. Disabling guidance at selected sampling steps also did not work, guidance seems to have a cumulative effect over multiple consecutive steps.

Questions

- The authors claim to estimate the optimal guidance interval in practice, the lower limit should be set to zero first to optimize the upper limit, then fix the upper limit and optimize the lower limit. However, they wrote in the introduction that limiting the guidance interval allows for a higher guidance weight. So it seems counterintuitive that one can have a constant weight but with difference sized intervals.

1.17 Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure

Xiang Li et al. (2024). "Understanding Generalizability of Diffusion Models Requires Rethinking the Hidden Gaussian Structure". In: *NeurIPS*

- In previous work, it was shown that diffusion models exhibit good generalization capacity. During training, recent research demonstrates that diffusion models operate in two distinct regimes: (i) a memorization regime, where models primarily reproduce training samples and (ii) a generalization regime, where models generate high-quality, novel images that extend beyond the training data. In the generalization regime, a particularly intriguing phenomenon is that diffusion models trained on non-overlapping datasets can generate nearly identical samples.
- The authors show that diffusion models have an inductive bias towards capturing and utilizing the Gaussian structure (covariance information)

of the training set. This inductive bias seems unique to diffusion models and becomes more apparent when the model's capacity is relatively small compared to the training data set size. In the overparameterized regime, this inductive bias emerges during initial training phases before the model fully memorizes the training data.

- The recently observed strong generalization results from diffusion models learning certain common low-dimensional structural features shared across non-overlapping datasets can be partially explained through the Gaussian structure.
- The observed Gaussian structure is the optimal solution to the denoising score matching objective under the constraint that the model is linear.
- Empirically by distilling models from a non-linear model, in the high-noise regime of training, only coarse image structures are generated. The linear, Gaussian, multi-delta and non-linear model counterparts behave very similar. In the low-noise regime, only subtle, imperceptible details are added to the generated images. Both linear and multivariate Gaussian models effectively approximate the non-linear model. Even the nonlinear model exhibits high linearity shown by approximating an identity function with 1 on the skip connections and 0 for the network weights. In the intermediate-noise regime, realistic image content is generated with significant nonlinearity.
- A linear diffusion model can still generate images that resemble those generated with a nonlinear model in terms of overall image structure and some details.
- Modeling $p_{\text{data}}(x)$ as a multi-delta distribution reveals a key insight: while unconstrained optimal denoisers perfectly capture the scores of the empirical distribution, they have no generalizability. In contrast, Gaussian denoisers, despite having higher score approximation errors due to the linear constraint, can generate novel images that closely match those produced by the actual diffusion models. This suggests that the generative power of diffusion models stems from the imperfect learning of the score functions of the empirical distribution.
- Although real-world image distributions are significantly different from Gaussian, the findings imply that diffusion models have the bias towards learning and utilizing low-dimensional data structures, such as the data covariance, for image generation. However, the underlying mechanism by which the nonlinear diffusion models, trained with gradient descent, exhibit such linearity remains unclear and warrants further investigation.
- The authors define a generalization score to measure how well a diffusion

model generalizes beyond the training data set:

$$\text{GL Score } := \frac{1}{k}\sum_{i=1}^{k}\frac{\|\boldsymbol{x}_i - \mathrm{NN}_Y\left(\boldsymbol{x}_i\right)\|_2}{\|\boldsymbol{x}_i\|_2}$$

Questions

- If it is true that during high-noise and low-noise regimes of diffusion sampling, the model shows high linearity, what does that mean for CFG? How does it behave differently? Is it connected to the CFG in the limited interval where CFG is only applied on the non-linear parts of sampling?
- What would happen if we were to apply the GL score to CFG? It should behave proportional to the guidance scale.

1.18 (skip) Understanding and improving training-free loss-based diffusion guidance

Yifei Shen et al. (2024). "Understanding and improving training-free loss-based diffusion guidance". In: *NeurIPS* Approximating $\nabla_{\boldsymbol{x}_t} \log p_t\left(\boldsymbol{y} \mid \boldsymbol{x}_t\right)$ for guidance in high-dimensional spaces requires solving an intractable integral or training a robust classifier. If a classifier is useful for guidance, it should be robust (aka low Lipschitz constant) to avoid adversarial guidance gradients. By employing (in practice 10) random augmentations to the one step denoised estimate improves classifier robustness. Using a Polyak step size (normalized gradient to balance the magnitude of the guidance and diffusion term) further improves guidance convergence.

1.19 (skip) CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling

Seyedmorteza Sadat et al. (2024). "CADS: Unleashing the Diversity of Diffusion Models through Condition-Annealed Sampling". In: *ICLR*

- High classifier guidance scale means sampling from a very peaked distribution, which leads to non-diverse samples (especially for the same conditioning)
- The diversity can be increased while maintaining high sample quality by adding scheduled, monotonically decreasing Gaussian noise to the conditioning vector during inference. This noising is interpreted as smoothing the conditional distribution, especially early in the sampling process.
- The schedule reduces the amount of Gaussian corruption as the inference processes, and a piecewise linear schedule for different noise/guidance weightings (similar to diffusion forward process). They also rescale the perturbed conditoning signal to retain the mean and std of the original conditioning signal.

1.20 Boosting Diffusion Models with Moving Average Sampling in Frequency Domain

Yurui Qian et al. (2024). "Boosting Diffusion Models with Moving Average Sampling in Frequency Domain". In: *CVPR*

- The authors propose to use moving averages over the x_0-prediction to stabilize sampling. For DDIM, this is straightforward plug-and-play. For other samplers, the x_0-prediction can be calculated by the noise prediction and then plugged in for sampling.
- Through the intuition that the diffusion model first generates low-frequency signals and then gradually adds high-frequency details, they employ a Discrete Wavelet Transform (DTW) to decompose the x_0-prediction into four wavelet subbands and separately apply moving averages with different parameters on each band. At each time step, the bands are converted back to the image domain via Inverse DWT.
- The method is only evaluated on FID, the diversity is not measured so the results are difficult to interpret.

1.21 Analysis of classifier-free guidance weight schedulers

Xi Wang et al. (2024a). "Analysis of classifier-free guidance weight schedulers". In: *TMLR*

- Finding the right balance between sample quality and diversity is a challenge in picking a suitable guidance scale ω
- Instead of a constant number, use a simple monotonically increasing schedule as too much guidance at the beginning of the denoising process is harmful. More advances schedules like parameterized ones can increase performance further but do not generalize across models and tasks.
- No guidance at the beginning of the sampling improves FID, no guidance at the end reduces FID. However, an adequate level of guidance across the entire sampling process is necessary for the best performance. The authors hypothesize that the guidance and generation terms may be adversarial during inference.
- Metrics: FID for sample quality, Clip-Score (CS) for alignment and the std over Dino-v2 image embeddings for diversity over multiple generations of the same prompt

1.22 (skip) The Unreasonable Effectiveness of Gaussian Score Approximation for Diffusion Models and its Applications

Binxu Wang and John Vastola (2024). "The Unreasonable Effectiveness of Gaussian Score Approximation for Diffusion Models and its Applications". In:

- In the high-noise regime of diffusion models, the score predicted by the neural network is well-approximated by a Gaussian model. In the low-noise regime, Gaussian mixture models approximate neural scores better than the score of the training distribution.
- The solution of the Gaussian model suggests a geometric sample generation: the estimated data point (x_0) begins at the center of the distribution and travels along the data manifold, moving first along high variance axes and then along the lower variance axes. Concurrently, the state x_t in the ambient space rotates towards the evolving endpoint estimate.

1.23 REG: Rectified Gradient Guidance for Conditional Diffusion Models

Zhengqi Gao et al. (2025). "REG: Rectified Gradient Guidance for Conditional Diffusion Models". In: *ICML*

- The authors claim that the currently employed CFG is only an approximating to the correct time-aware CFG, which would in practice prohibitively require sampling x_0 for each x_t.
- The paper proposes a rectified guidance term. Despite lower error and better FID/IS pareto fronts, it requires computation of an additional Jacobian matrix compared to vanilla CFG.

1.24 Eliminating Oversaturation and Artifacts of High Guidance Scales in Diffusion Models

Seyedmorteza Sadat et al. (2025a). "Eliminating oversaturation and artifacts of high guidance scales in diffusion models". In: *ICLR*

- The authors disentangle the CFG update term into parallel and orthogonal components and show that the parallel component primarily causes oversaturation, while the orthogonal component enhances image quality. The parallel component can be seen as a simple "gain", pushing the values towards extremes of their range.
- Rescaling and momentum is added to the CFG updates by the gradient ascent analogy. The CFG update rule can be interpreted as stochastic gradient ascent on the ℓ_2 distance between the conditional and unconditional prediction with a learning rate of $w - 1$.
- If the delta between the conditional and unconditional predictions is too large, the update can cause drifts in the sampling process. They propose to constrain the updates within a sphere so that the delta stays closer to the conditional prediction.

- The momentum (average pixel values of previous updates) is used in a repulsive way to down-weigh components already present in previous sampling steps. They call it reverse momentum.
- Adaptive projected guidance (APG) retains the quality-boosting advantages of CFG while preventing oversaturation at higher guidance scales. The authors claim APG also increases diversity. APG notably works over the denoised predictions instead of the noise predictions, even though they can be computed interchangeably.
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Questions

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Overall Summary and Questions

- CFG is a technique to mix scores of a conditional and unconditional model to balance mode coverage and sample fidelity. This combined model is strangely better than a conditional or an unconditional model. In practice, the conditional model gets the majority of the compute budget and its score is most influential during sampling (high weight).
- CFG was originally intended to construct a new noise network to sample from a scaled distribution instead. This has been shown to not be the case for simpler tasks such as Gaussian mixtures. The intermediate distributions and the final distributions are very difficult to characterize outside specifically constructed cases (multi-delta, Gaussian, linear model).
- CFG provides a way to trade-off mode coverage (prompt alignment, diversity) with sample fidelity, although empirically it has been shown that very high prompt alignment and very high sample fidelity can be achieved while maintaining high generalization capability and only sacrificing some diversity in generations. Why and how this happens is still unclear.
- One intuition why CFG works so well could be that we can not train, but we can sample from a higher quality underlying data distribution. Natural images contain for example many objects and are often cluttered, whereas there is a subset of clean images with a single object which is perceived as higher quality. It is however questionable how to measure the higher perceived quality (FID compares to samples from the distribution).
- Since people are maximizing the FID and IS scores as this is measurable, are there optimal mixtures of the conditional / unconditional scores such that each metric is optimized?

- CG takes an unconditional model and includes a gradient ascent step towards a classifier. CFG largely takes a conditional model and removes "unconditional likelihood". Could we go back and do the same for CG, so train a conditional model and then during sampling increase diversity through increasing classifier entropy?

TODOs

- Marta Skreta et al. (2025b). "The Superposition of Diffusion Models Using the It\^ o Density Estimator". In: *ICLR*
- Seyedmorteza Sadat et al. (2025b). "No training, no problem: Rethinking classifier-free guidance for diffusion models". In: *ICLR*
- Chubin Chen et al. (2025). "S^2-Guidance: Stochastic Self Guidance for Training-Free Enhancement of Diffusion Models". In: *arXiv 2508.12880*
- Krunoslav Lehman Pavasovic et al. (2025). "Classifier-Free Guidance: From High-Dimensional Analysis to Generalized Guidance Forms". In: *arXiv*
- Jie Shao et al. (2025). "Images Speak Louder Than Scores: Failure Mode Escape for Enhancing Generative Quality". In: *arXiv:2508.09598*
- Marta Skreta et al. (2025a). "Feynman-kac correctors in diffusion: Annealing, guidance, and product of experts". In: *ICML*
- Peng Sun et al. (2025a). "Unified Continuous Generative Models". In: *arXiv:2505.07447*
- Valentin De Bortoli et al. (2025). "Distributional diffusion models with scoring rules". In: *ICML*
- Qiao Sun et al. (2025b). "Is Noise Conditioning Necessary for Denoising Generative Models?" In: *arXiv:2502.13129*
- Zhicong Tang et al. (2025). "Diffusion models without classifier-free guidance". In: *arXiv:2502.12154*
- Zhen Yang et al. (2025). "Efficient Training-Free High-Resolution Synthesis with Energy Rectification in Diffusion Models". In: *arXiv:2503.02537*
- Xiaoming Zhao and Alexander G Schwing (2025). "Studying Classifier (-Free) Guidance From a Classifier-Centric Perspective". In: *arXiv:2503.10638*
- Rafał Karczewski et al. (2025). "Devil is in the details: Density guidance for detail-aware generation with flow models". In: *ICML*
- Kulin Shah et al. (2025). "Does generation require memorization? creative diffusion models using ambient diffusion". In: *ICML*

2 Conditional Diffusion (Image) Sampling Metrics

2.1 A note on the evaluation of generative models

Lucas Theis et al. (2016). "A note on the evaluation of generative models". In: *ICLR*

- The objective function in training yields a different behavior of the model. Optimizing maximum mean discrepancy (MMD) or Jensen-Shannon divergence (JSD) yields a Gaussian which fits one mode well but ignores other parts of the data. Maximizing average log-likelihood / minimizing the Kullback-Leibler divergence avoids assigning extremely small probability to any data point but assigns a lot of probability mass to non-data regions. Depending on the downstream task, a different property may be desired.

- For evaluation, the average log-likelihood is disconnected to image quality in high-dimensional spaces:

  - Poor log-likelihood but great samples:
    simple lookup table for training data can generate great samples but has poor likelihood on unseen data, a mixture of Gaussians centered around training data can generate high quality images for imperceptible noise but has poor likelihood estimation

  - Great log-likelihood but poor samples:
    consider a mixture model of $0.01p(x) + 0.99q(x)$, where $p(x)$ is a good model wrt to log-likelihood and $q(x)$ is a bad model. Therefore, 99% of samples come from the bad model. However, $\log[0.01p(x) + 0.99q(x)] \geq \log[0.01p(x)] = \log p(x) - \log 100$, where $\log p(x)$ is proportional to the dimensionality d but $\log 100$ stays constant. The difference between log-likelihoods of different models on 32x32 data can already be in the thousands, wheres $\log 100$ is only about 4.61 nats. This shows that a model can have large average log-likelihood but generate very poor samples.

  - Good log-likelihood and great samples:
    in the above mixture, the good model could have been chosen 99% of the time, producing samples indistinguishable from the real images 99% of the time. The log-likelihood would however only change by at most 4.61 nats. This shows that any model can be turned into a model which produces realistic samples at little expense to its log-likelihood. Log-likelihood and visual appearance of samples are therefore largely independent.

- Evaluation based on samples and nearest neighbor in the data set is difficult:

  - Nearest neighbors are typically determined based on Euclidean distance. But already perceptually small changes can lead to large changes in Euclidean distance. A simple pixel shift can lead to perceptually negligible difference but large Euclidean distance, possibly changing the nearest neighbor. Note that with a bigger dataset, a switch to a different nearest neighbor becomes more likely.
  - Even when overfitting, most models will not reproduce perfect or trivially transformed copies of the training data. In this case, no distance metric will find a close match in the training set.

- An evaluation based on samples is biased towards models which overfit and therefore a poor indicator of a good density model in a log-likelihood sense, which favors models with large entropy. Conversely, a high likelihood does not guarantee visually pleasing samples. The optimal metric for image quality would be human visual assessment.

2.2 Inception Score

Tim Salimans et al. (2016). "Improved techniques for training gans". In: *NeurIPS*

- A test of human visual quality assessment to distinguish real and generated images shows that: a) the metric depends on the task setup and the motivation of the annotators, and b) the results change drastically when the annotators get feedback about their mistakes and learn from it. The goal is to have an automated metric, which however correlates well with human judgement.

- Generated samples are fed into an inception model that was trained on ImageNet to get the conditional label distribution $p(y|x)$. Images that contain meaningful objects should have a conditional label distribution $p(y|x)$ with low entropy (they belong to few object classes). The model should also generate varied images, so the marginal $p(y|x = G(z))dz$ should have high entropy (the variance over the images should be large). Combining these two requirements, the proposed metric is:

$$IS\left(p_{gen}, p_{dis}\right) := \exp\left(\mathbb{E}_{x\sim p_{gen}}\left[D_{KL}\left(p_{dis}(\cdot \mid x) \| \int p_{dis}(\cdot \mid x) p_{gen}(x) dx\right)\right]\right) \quad ,$$

  with p_{gen} as the generative distribution and p_{dis} as an inceptionv3 classifier predicting a lable distribution per sample. The metric has to be calculated over a large sample set (50k) as it measures diversity as well.

2.3 Frenchet Inception Distance

Martin Heusel et al. (2017). "Gans trained by a two time-scale update rule converge to a local nash equilibrium". In: *NeurIPS*

- Any distance between the probability of observing real world data p_{data} and the probability of generating model data p_{gen} can serve as a performance measure for generative models.

- Likelihood is the best known measure, but heavily depends on the noise assumptions for the real data and can be dominated by single samples.

- FID proposed as improvement over IS, as extracted statistics of generated samples are not compared to statistics of the real world samples.

- The last pooling layer in a pre-trained inception model is used to extract vision-relevant features (d=2048). Then, (for practical reasons) the first and second moments are considered. As the Gaussian is the maximum entropy distribution for given mean and covariance, the assumption is that the extracted features follow a multidimensional Gaussian. The difference between the Gaussian of the generated samples and the Gaussian of the real data is measured by the Frenchet Distance (aka Wasserstein-2 distance) as:

  $$FID = \|\mu - \mu_w\|^2 + \mathrm{tr}\left(\Sigma + \Sigma_w - 2\left(\Sigma\Sigma_w\right)^{\frac{1}{2}}\right) \quad ,$$

  where $\mathcal{N}(\mu_w, \Sigma_w)$ is a multivariate normal distribution estimated from inceptionv3 features of 50k fake images, and $\mathcal{N}(\mu, \Sigma)$ as a multivariate normal distribution estimated from inceptionv3 features of all real images in the training data set.

COMMENT Philipp: Are FID and IS really that different? The difference between the features in the last pooling layer and the output distribution should be almost none (`https://pytorch.org/hub/pytorch_vision_inception_v3/`). They do in the end measure different things, but there may be a theoretical equivalence under assumptions?

2.4 Precision & Recall

Mehdi SM Sajjadi et al. (2018). "Assessing generative models via precision and recall". In: *NeurIPS*

- FID correlates well with perceived quality and is sensitive to mode dropping, but it can't distinguish the quality of generated samples from the coverage of the target distribution. Failure cases are also hidden in this single quantity.

- Given a reference distribution P and a learned distribution Q, precision intuitively measures the quality of samples from Q, while recall measures the proportion of P that is covered by Q. These measures can quantify the degree of mode dropping and mode inventing based on samples from the true and the learned distributions as well. (The measure shows that GANs often produce "sharper" images, but can suffer from mode collapse (high precision, low recall), while VAEs produce ”blurry” images, but cover more modes of the distribution (low precision, high recall))

- A pre-trained Inception network embeds the samples and real images (for example last pooling layer). The embeddings are clustered by mini-batch k-means with k = 20. This reduces the problem to a one dimensional problem where the histogram over the cluster assignments can be meaningfully compared. Hence, failing to produce samples from a cluster with many samples from the true distribution will hurt recall, and producing samples in clusters without many real samples will hurt precision. As the clustering algorithm is randomized, we run the procedure several times and average over the PRD curves.

2.5 Improved Precision & Recall

Tuomas Kynkäänniemi et al. (2019). “Improved precision and recall metric for assessing generative models”. In: *NeurIPS*

- ”For instance, it is interesting that while recent state-of-the-art generative methods claim to optimize FID, in the end the (uncurated) results are almost always produced using another model that explicitly sacrifices variation, and often FID, in favor of higher quality samples from a truncated subset of the domain.”

- The classic definition of precision and recall is sufficient for disentangling the effects of sample quality and manifold coverage. However, it fails if many samples are packed together (for example as a result mode collapse or truncation). It also does not allow binary membership queries such as “does sample x lie in the support of distribution P ?”.

- Take an equal number of real and generated samples and embed them into a high-dimensional feature space of a pre-trained classifier. For both sets of images, the manifold is estimated by calculating pairwise Euclidean distances between all feature vectors in the set and, for each feature vector, forming a hypersphere with radius equal to the distance to its kth nearest neighbor. Together, these hyperspheres define a volume in the feature space that serves as an estimate of the true manifold.

- Precision is quantified by querying for each generated image whether the image is within the estimated manifold of real images. Recall is calculated by querying for each real image whether the image is within estimated manifold of generated image.

- Like FID, the improved precision/recall needs a large sample size (50k). The size of the neighborhood, k, is a compromise between covering the entire manifold (large values) and overestimating its volume as little as possible (small values). In practice, $k = 3$ is a robust choice.

2.6 Density & Coverage

Muhammad Ferjad Naeem et al. (2020). “Reliable Fidelity and Diversity Metrics for Generative Models”. In: *ICML*

- The precision and recall metrics are already a big improvement over FID, but are still not reliable yet. For example, they fail to detect the match between two identical distributions, they are not robust against outliers, and the evaluation hyperparameters are selected arbitrarily. Even the improved Precision/Recall fall short on the requirements.
- Pre-trained feature extractors have a strong data set bias and fail if the target data distribution differs from the trained data statistics. Therefore, random CNN feature embeddings can be used in such cases.
- Original Precision/Recall assumes that the embedding space is uniformly dense, relies on the initialization-sensitive k-means algorithm for support estimation, and produces an infinite number of values as the metric.
- Improved Precision/Recall is still vulnerable to outliers and limited by computational efficiency. Generating many fake samples around a real outlier is enough to increase the precision measure. The recall measure gains points for real samples contained in the overestimated fake manifold. The manifold computation is per model and expensive.
- Instead of precision quantifying a binary decision if a generated data point is within any data point hypersphere, density counts how many data set spheres contain the generated image. Density therefore rewards generating samples in high density regions, which improves robustness towards outliers.
- Instead of constructing the manifold around fake samples and quantifying how many real images are within the generated manifold, coverage estimates the manifold around the real images and measures the fraction of real samples whose neighborhood contain at least one fake sample. This requires the manifold computation only once per data set (instead of model), and improves the robustness towards outliers as the generated samples tend to have more outliers than the data.

2.7 Consistency-diversity-realism Pareto fronts of conditional image generative models

Pietro Astolfi et al. (2024). “Consistency-diversity-realism Pareto fronts of conditional image generative models”. In: *arXiv:2406.10429*

- Current research in generative models mostly focuses on creative applications that are predominantly concerned with human preferences of image quality and aesthetics (a single high-quality and consistent sample fulfills the current optimization criteria. This vastly disregards representation diversity).

- Pareto fronts provide a holistic view on the consistency-diversity-realism multi-objective. Inter-sample similarity and recall is used to measure representation diversity; image reconstruction quality and precision to quantify realism; and the Davidsonian scene graph score to assess prompt-generation consistency. The metrics are split for marginal metrics (metric over all generations and all prompts) and conditional metrics (metric over all generations per prompt, can be averaged for a summary statistic).
- The experiments suggest that realism and consistency can both be improved simultaneously; however there exists a clear tradeoff between realism/consistency and diversity. By looking at Pareto optimal points, earlier models are better at representation diversity and worse in consistency/realism, and more recent models excel in consistency/realism while decreasing significantly the representation diversity.
- CFG is a inference trade-off between image fidelity and diverstiy, post hoc filtering is used to improve consistency

3 Diffusion Memorization

3.1 Do deep generative models know what they don't know?

Eric Nalisnick et al. (2019). "Do deep generative models know what they don't know?" In: *ICLR*

- Discriminative models using NNs can predict the wrong class with high confidence (for example by rotating MNIST digits). One way to avoid this is to only predict the class on inputs with high density under a trained generative model.
- Autoregressive models, flow-based models and VAEs assign higher density to SVHN than to CIFAR-10 as their training data. This observation translates to other image data sets.
- Through a second order analysis, the difference in likelihoods on the training data set and a secondary data set can be explained by the model curvature and the data's second moment.
- SVHN seems to "sit inside of" CIFAR-10 with roughly the same mean but smaller variance, resulting in its higher likelihood.
- The difference of likelhoods scales with the difference in data variances. For example, SVHN has the smallest variance and largest likelihood. This means the likelihood can be arbitrarily increases by decreasing the data set variance (which means graying RGB images, so bringing the pixel values closer to 128). Reducing the variance of latent representations has the same effect.

- Comparing the likelihoods of deep generative models alone cannot identify the training set or inputs like it.

3.2 Does learning require memorization? a short tale about a long tail

Vitaly Feldman (2020). "Does learning require memorization? a short tale about a long tail". In: *ACM STOC*

- Setting is discriminative models - not generative models
- The only way to fit an example whose label cannot be predicted based on the rest of the dataset is to effectively memorize it. If a model has to fit random labels and the data set is thus not informative about the label of a data point, the model has to memorize it.
- An explanation why memorization is necessary for close-to-optimal generalization error is not the noise inherent in the labels but rather an insufficient amount of data to predict accurately on rare and atypical instances. Such instances are usually referred in practice as the "long tail" of the data distribution.
- There is evidence that the examples on which a differentially private model errs are either outliers or atypical examples. The reason why learning with DP cannot achieve the same accuracy as non-private learning is that it cannot memorize the tail of the mixture distribution. This view also explains the recent empirical results showing that the decrease in accuracy is larger for less well represented subpopulations.
- For typical image and text datasets and learning algorithms, there is a significant fraction of examples whose memorization is necessary for predicting accurately on examples from the same subpopulation in the test set. This has been confirmed on MNIST, CIFAR-10/100, SVHN and ImageNet datasets revealing numerous visually similar pairs of relatively atypical examples.

3.3 On Memorization in Probabilistic Deep Generative Models

Gerrit van den Burg and Chris Williams (2021). "On memorization in probabilistic deep generative models". In: *NeurIPS*

- A common technique to measure memorization of a sample in deep generative models is a nearest neighbor search in the training set. As euclidean distance in pixel space is easily fooled (for example by pixel shifts), NN in the feature space of a secondary model or cropping/downsampling before NN can be used.

- The memorization aspect for the paper is ”an increased probability of generating a sample that closely resembles the training data in regions of the input space where the algorithm has not seen sufficient observations to enable generalization” (for example if an outlier is responsible for all the density in the area, a generated sample out of this area should be very similar to the outlier data point). This is in contrast to memorization due to data duplication, which is the case for common data sets.

- The leave-one-out (LOO) memorization score is proposed based on the change in the probability density of an observation that occurs when it is removed from the data. It is defined for a single data point as the difference in log probability for a model trained on the entire data set versus a model trained on the data set without the data point. Each log probability calculation is averaged over L training runs with different initializations.

- In practice to make the score tractable, a K-fold approach is used instead, where disjoint subsets of the training set are removed and the log probability for all observations is computed ($K = 10$, $L = 10$).

- While the LLO memorization score directly measures how much the model relies on the presence of a data point for the local probability density, nearest neighbor methods test how “close” samples from the model are to the training or validation data. They thus require a meaningful distance metric (which is non-trivial for high-dimensional data) and are subject to variability in the sample and validation sets. Nearest neighbor examples and hypothesis tests can be informative and may detect global memorization, but to understand memorization at an instance level, the LLO memorization score is to be preferred.

- For a differentially private model, it is guaranteed to have an LLO memorization score bounded by the degree of privacy (the converse is not true).

- Besides DP training, another way of limiting memorization is to include a model component with broad support but low probability (such as a Gaussian with high variance). This ensures the log probability for atypical observations to be small whether they are included in the training data or not, resulting in a low memorization score.

3.4 On the De-duplication of LAION-2B

Ryan Webster et al. (2023). “On the de-duplication of laion-2b”. In: *arXiv:2303.12733*

- LAION-2B is an extremely large image database containing 2 billion images, which was the basis for breakthroughs in computer vision.

- Duplicate training data points pose degrade model utility through issues such as higher memorization. This is especially true for large scale data sets thorugh web scrapers, potentially containing many duplicates.

- The paper proposes an algorithm to efficiently find duplicates at scale, identifying over 700 million (1/3) of the LAION-2B images as duplicates. The focus of the method is scalability rather than perfect duplicate detection.

- The algorithm is called Subset Nearest Neighbor CLIP compression (SNIP) and builds on top of pre-trained CLIP network. The text and image features are extracted through a CLIP encoder. The SNIP algorithm has two parts: a "latent" clip loss between image and text encodings, and a clip loss between the clip encoding of an image and that of the ℓ_2 nearest neighbor in the pixel space.

- The nearest neighborhood computation is sped up by computing pairwise distances in chunks of the input data only (each chunk is 10m data points). To further speed up the NN search, they use an inverted file system. First one computes the k-means centroids of the database vectors and then groups vectors to their closest centroid to form the inverted files. During search, queries only look through the closest inverted files by exhaustive search over the closest centroids, which is tractable because the number of centroids is typically small.

3.5 Extracting Training Data from Diffusion Models

Nicolas Carlini et al. (2023). "Extracting training data from diffusion models". In: *USENIX Security*

- A common question for largescale generative models is whether their impressive results arise from truly novel generations, or are instead the result of direct copying and remixing of their training data. Feldman 2020's theory that memorization is necessary for generalization in classifiers may extend to generative models, raising the question of whether the improved performance of diffusion models compared to prior approaches is precisely because diffusion models memorize more.

- From model attack literature, there is a difference between a black-box adversary who can only provide prompts or a white-box adversary who can also control the models randomness. There is a further distinction between attack strength (the strongest first): data extraction (sample an image almost identical to one of the training set), data reconstruction (recover a full training image based on a known part of the image), and membership inference (confirming if an image was part of the the training set)

- (k,ℓ,δ)-Eidetic Memorization is a notion of approximate memorization. An example x is memorized by a diffusion model if x is extractable from the diffusion model, and there are at most k training examples that are closer to x than some threshold. In practice, the distance is commonly an ℓ_2 distance normalized by the dimensions ($\ell \in [0, 1]$). The distance can be the cosine similarity in an CLIP embedding space or an ℓ_2 distance over non-overlapping image patches.

- Higher capacity models, more training iterations, duplicates in the training set, outliers in the training set and a smaller data set are potential contributors to more memorization.
- Diffusion models are significantly less private than classifiers trained on the same data, it is however not clear to what extend this is because they are often trained far longer than classifiers.
- As the quality of generative models increases (FID) so too does the privacy leakage. These results are concerning because they suggest that stronger models of the future may be even less private.
- Diffusion models are less private than GANs for membership inference attacks under default training settings, even when the GAN attack is strengthened due to having access to the discriminator (which would be unlikely in practice, as only the generator is necessary to create new images).
- Diffusion models memorize more data than GANs, even when the GANs reach similar performance. Diffusion models and GANs memorize many of the same images. Many examples that are easy to extract are duplicated many times (e.g., > 100) in the training data.

3.6 Diffusion Art or Digital Forgery? Investigating Data Replication in Diffusion Models

Gowthami Somepalli et al. (2023a). "Diffusion art or digital forgery? investigating data replication in diffusion models". In: *CVPR*

- Training data duplication and text conditioning play significant roles in the memorization of diffusion models from empirical studies
- Definition of replication: A generated image has replicated content if it contains an object (either in the foreground or background) that appears identically in a training image, neglecting minor variations in appearance that could result from data augmentation.
- Similarity between images: It is common to compare two images via the inner product of feature vectors (either the [CLS] token or average-pooled representations). Inner product metrics measure global, rather than local similarity. This is because inner product spaces are metric spaces and thus satisfy the triangle inequality. For example, if image A has visual similarity to images B and C, yet both of these images are not similar, the triangle inequality $d(B, C) \leq d(A, B) + d(A, C)$ would force B and C to be close as well (even though they are not similar).

- To circumvent this, they propose a split-product metric that breaks each feature vector into chunks, computes inner products between corresponding chunks, and returns the maximum across these inner products.
- The split product metric is a more semantic measure of similarity. As expected, the inner product metrics enforces a stricter notion of pixel-wise similarity.
- Diffusion models trained on smaller datasets tend to generate images that are copied from the training data. The amount of replication reduces with a larger size of the training set. It seems that replicated content tends to be from training images that are duplicated more than a typical image.
- There is a relationship between the intra-class diversity and similarity scores: classes with higher self-similarity scores on average have higher maximum similarity score amongst matches with generated samples.

3.7 Understanding and mitigating copying in diffusion models

Gowthami Somepalli et al. (2023b). "Understanding and mitigating copying in diffusion models". In: *NeurIPS*

- While it is widely believed that duplicated images in the training set are responsible for content replication at inference time, the text conditioning of the model plays a similarly important role. Replication can happen even when the duplication rate in the training set is greatly reduced. Data replication often does not happen for unconditional models, while it is common in the text-conditional case.
- When training a diffusion model, it is crucial to carefully manage data duplication to strike a balance between reducing memorization and ensuring high-fidelity of generated images. However, controlling duplication is not enough to prevent replication of training data.
- A model is more likely to memorize images when the captions are more unique or have high perplexity. However, the model does not exhibit the highest level of memorization when using "random captions", meaning that captions should also be correlated with image content in order to help the model retrieve an image from its memory maximally. Caption specificity is closely tied to data duplication. Highly specific (or high perplexity) captions act like keys into the memory of the diffusion model that retrieve specific data points.
- Compared to full duplication (image + caption), partial duplication (image + different semantically similar captions) substantially mitigates copying behavior, even as duplication rates increase.

- A larger and more diverse training set allows the model to generalize better and produce more varied outputs. Choosing the optimal length of training, quantity of training data, and training setup is a trade-off between model quality and memorization. When images do get memorized in the absence of training data duplication, they are likely to be "simple" in structure.
- Memorization can decrease even when duplication increases, as long as captions are sufficiently diverse.
- Before training memorization strategies can involve resampling captions to avoid duplicates and removing images with limited complexity.

3.8 Diffusion Probabilistic Models Generalize when They Fail to Memorize

TaeHo Yoon et al. (2023). "Diffusion Probabilistic Models Generalize when They Fail to Memorize". In: *SPIGM Workshop - ICML*

- They define learning to include the following two types: rote learning (i.e., memorization) and the conceptual learning (which enables generalization). A generative model performs perfect rote learning when it learns to generate from a delta distribution around each training data point. In this case, we say the model has perfectly memorized the training set and say the generated samples are replicates of the training data. A generative model performs perfect conceptual learning when it learns to generate from the true underlying distribution generating the training data. In this case, we say the model generalizes, and this is the desired behavior of a generative model.
- For deep supervised learning, trained deep neural networks generalize well despite memorizing (interpolating) training data where "benign overfitting" (Belkin, 2021), claims that overparameterization is not detrimental to the generalization performance.
- DPMs generalize when they are underparameterized, which is consistent with the classical statistical view that overfitting should be avoided. The nature of learning of DPMs (and perhaps for generative models as well) is different from that of modern deep supervised learning.
- As a replication detection, they use the eidetic ℓ_2 distance from Carlini et al. 2023. They mark a sample as replicated if the ℓ_2 distance to the nearest neighbor in the training set is less than 1/3 of the distance to the second nearest neighbor.
- Diffusion models performs rote learning up to certain point, and then display an exponential decay trend in replicating behavior with respect to increasing train set size. Conceptual learning is the result of failing to brute-force-memorize, given that the train set possesses enough complexity to enable generalization.

- A model may perform different types of (rote or conceptual) learning over different classes. They observe that for a minority class, the learning pattern is closer to rote learning. On the other hand, the same model tends to perform conceptual learning with respect to a majority class.

3.9 Towards memorization-free diffusion models

Chen Chen et al. (2024). "Towards memorization-free diffusion models". In: *CVPR*

- The primary causes of memorization is image and text duplication in training datasets, along with the high specificity of user prompts for text conditioning.
- Pixel-level memorization is detected by the negative L2 norm in pixel space of a sample and its NN in the training data, normalized by 1/2 of the average L2 norm to the 50 NN in the training data.
- Object-level memorization is detected by the dot product of Self-supervised Copy Detection (SSCD) embeddings between the sample and the train NN
- Anti-Memorization Guidance (AMG) builds on top of CFG and CG to prevent memorization during sampling when there is high similarity of the one-step denoising estimate to any training data point, consisting of three terms:
  - Desspecification guidance reduces caption specificity by guiding in the opposite direction of CFG, bounded by scaling to prevent low text-image alignment
  - Caption deduplication guidance steers away from the direction of perfect text conditioning simulated through duplicated captions that would lead to memorization.
  - Dissimilarity guidance guides towards low similarity score wrt to the closest training data point.

3.10 Detecting, Explaining, and Mitigating Memorization in Diffusion Models

Yuxin Wen et al. (2024). "Detecting, explaining, and mitigating memorization in diffusion models". In: *ICLR*

- Novel approach to detect memorized prompts and mitigate memorization for Stable Diffusion

3.11 Finding NeMo : Localizing Neurons Responsible For Memorization in Diffusion Models

Dominik Hintersdorf et al. (2024). "Finding nemo: Localizing neurons responsible for memorization in diffusion models". In: *NeurIPS*

- Memorization in text-to-image diffusion models is localized to specific neurons.

3.12 The Emergence of Reproducibility and Consistency in Diffusion Models

Huijie Zhang et al. (2024). "The emergence of reproducibility and consistency in diffusion models". In: *ICML*

- Measure of model reproducibility: the probability (empirical estimate based on 10k noise samples) of a generated sample pair (x1, x2) from two different diffusion models to have self-supervised copy detection (SSCD) similarity larger than 0.6 starting from the same iid noise; the probability of an MAE < 15.0 based on [0,255] image comparisons

- Measure of model generalizability: 1 - probability of maximum model reproducibility score over the training dataset larger than 0.6. This measures the dissimilarity between the generated sample and all samples from the training dataset.

- "Consistent model reproducibility": Given the same noise input, different diffusion models often generate remarkably similar samples through a deterministic sampler, manifesting in memorization and generalization regimes. This phenomenon also occurs for conditional diffusion models and diffusion models for inverse modeling.

- Different diffusion models consistently reach the same data distribution and score function regardless of diffusion model frameworks, model architectures, training procedures, sampling procedures or perturbation kernels. This does not seem to hold for other generative models like GANs or VAEs.

- Diffusion models learn distinct distributions based on the training data size in two distinct training regimes: (i) "memorization regime," where the diffusion model overfits to the training data distribution, and (ii) "generalization regime," where the model learns the underlying data distribution

- Memorization regime occurs when the model has much larger capacity than the size of training data. The emergence of reproducibility is due to the fact that all diffusion models memorize the same multi-delta distribution of training samples. Given no generalizability, training diffusion models in this regime might hold limited practical interest.

- Generalization regime emerges when the diffusion model not only regains its reproducibility but also becomes capable of generating new samples distinct from the training data and usually happens when the diffusion model is trained on large dataset without full capacity to memorize the whole dataset. There is a clear transition from the memorization regime to the generalization regime as the training samples increase.
- Model reproducibility of conditional models exhibits in a structured way and is strongly related to unconditional counterparts. It appears that the conditional diffusion model learns to map from the same noise space to each individual image manifold corresponding to each class. In contrast, the unconditional diffusion model maps the noise space to a broader image manifold as a union over image class manifolds. A theoretical analysis of this unique reproducibility relationship holds the promise of providing valuable insights. When comparing the same model with identical initial noise but different class conditions, there is similarity in low-level structural attributes, such as color, despite differing in semantics.
- For solving inverse problem using diffusion models, model reproducibility holds only within the same type of network architectures.
- (strong claim) "First empirical evidence that diffusion models can overcome the curse of dimensionality when learning the underlying distribution, enabling effective generalization even with a limited number of training samples."
- Based on the findings of the paper, one can replicate the mapping from a trained diffusion model by training a new score function from generated data (through an open-source API).

3.13 An analytic theory of creativity in convolutional diffusion models

Mason Kamb and Surya Ganguli (2025). "An analytic theory of creativity in convolutional diffusion models". In: *ICML*

- Two inductive biases (locality and equivariance) in common architectures of diffusion models prevent convergence to the ideal score function. These two inductive biases do not fully capture the behavior of trained models, particularly those with self-attention architectures.

3.14 Selective Underfitting in Diffusion Models

Kiwhan Song et al. (2025). "Selective Underfitting in Diffusion Models". In: *arXiv:2510.01378*

3.15 How Diffusion models memorize

Juyeop Kim et al. (2025). "How Diffusion Models Memorize". In: *arXiv:2509.25705*

3.16 On Memorization in Diffusion Models

Xiangming Gu et al. (2025). "On Memorization in Diffusion Models". In: *TMLR*

- Assuming sufficient model capacity, diffusion models optimized by denoising score matching have an optimal solution, which can only produce samples that replicate the training data. This theoretical memorization behavior is empirically contradicted by the good generalization capability of SODA diffusion models like EDM.
- Memorization definition (Yoon et al. 2023): a sample is memorized if the ℓ_2 distance to the nearest neighbor is smaller than 1/3 of the distance to the second-nearest neighbor in the training data (empirical threshold aligned with human memorization)
- Most experiments are on CIFAR10 in a highly over-parameterized setting (maximum 5k images vs 56M parameter models), no memorization in unconditional setting with full 50k CIFAR10 -> conclusions are therefore not reliable
- Interestingly, an EDM model conditioned on a unique class label for CIFAR10 (50k images) memorizes more of the training data for more training epochs (eventually over 65%), whereas the unconditional model does not memorize at all
- A discriminative InceptionV3 (<25M parameters) can almost memorize ImageNet (1.28M images) (Zhang et al. 2017), a generative EDM (56M) can not memorize CIFAR10 (50K images). Over-parameterized models generalize well to real data distribution and even perfectly fit to the training dataset (benign overfitting). Nevertheless, diffusion models demonstrate adverse generalization performance.

3.17 On the Edge of Memorization in Diffusion Models

Sam Buchanan et al. (2025). "On the Edge of Memorization in Diffusion Models". In: *arXiv:2508.17689*

3.18 Other relevant papers

- Chiyuan Zhang et al. (2017). "Understanding deep learning requires rethinking generalization". In: *ICLR*
  - InceptionV3 (<25M parameters) can almost memorize ImageNet (1.28M images).
- Peter L Bartlett et al. (2020). "Benign overfitting in linear regression". In: *PNAS*
  - Over-parameterized models generalize well to real data distribution and even perfectly fit to the training dataset.

- Preetum Nakkiran et al. (2020). "Deep double descent: Where bigger models and more data hurt". In: *ICLR*